SafeInvest: Horizon-Aware Allocation with Cost-Gated Portfolio Updates

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Abstract

SafeInvest must choose suitable investments, avoid unnecessary trading, and recognize when automated advice should stop. We develop three horizon-specific portfolios using a convex expected-log-growth problem constrained by tail log drawdown. A separate execution rule moves partially toward the strategic allocation only when the estimated benefit exceeds transaction costs and a safety margin. Suitability is assessed at each customer's actual horizon using out-of-fold loss outcomes, conservative probability bounds, and a doubled-cost stress test. The selected, frozen Plan A assigns customers B and C to Conservative, E to Balanced, and A and D to human review.

All development uses 120 complete scenarios. A single frozen evaluation uses the remaining 60; its saved outputs provide the evidence reported here. At 30 years and a 0.25% transaction cost, Balanced A delivers a median real annual growth of 4.15%, compared with 3.38% for annual equal-weight rebalancing. Its 90th-percentile maximum drawdown is also higher: 48.02% versus 39.53%. Against annual rebalancing to the same strategic allocation, the execution rule reduces mean real fees by 26.83% and trades at 35.34% of subsequent annual reviews. A severe scenario nevertheless loses 65.19% from peak to trough. The evidence supports selective, explainable automation with explicit escalation; it does not establish a guarantee against loss or universal superiority over the baseline.

1 Decision Problem and Evidence

The advisor invests in equities, government bonds, and Treasury bills. Its responsibilities are linked: define Conservative, Balanced, and Growth products; match customers to a product or a human advisor; and choose among keeping holdings, rebalancing, changing the current target, and requesting human review. Optimizing returns alone does not address these service decisions.

The supplied data comprise 180 independent synthetic scenarios, each with 30 annual observations of asset returns and inflation. They are based on selected US series from the JST Macrohistory Database, Release 6, as described in the problem statement [1, 4]. Scenario numbers identify simulated paths, not historical episodes. We retain each complete path as one statistical unit and use the first H periods when evaluating a customer with horizon H.

Scenarios 1–120 form the development partition; scenarios 121–180 are reserved for the final evaluation. The latter was executed once on 17 October 2026 after model settings, customer assignments, and comparators were frozen. The final decision retains calibrated Plan A because subsequent development candidates did not satisfy the complete replacement criteria. Neural-network experiments are therefore outside the submitted strategy. Report preparation uses saved summaries and verification records only.

Our design separates three questions: what long-run allocation is appropriate, whether the customer can be served automatically, and whether an immediate trade is worth its cost. This makes the recommendation auditable and permits a justified refusal to automate.

2 Accounting, Assumptions, and Performance Measures

2.1 A causal, self-contained investment account

Initial nominal wealth is 100, held entirely in bills. Portfolios are long-only and fully invested: $w_i \geq 0$ and $\sum_i w_i = 1$. There are no contributions, withdrawals, leverage, taxes, or other fees. Annual decisions use only information available before that year's returns. Initial investment incurs a transaction fee; there is no assumed terminal liquidation. Bills provide liquidity but are not guaranteed to preserve purchasing power.

Let $h_{i,t}^-$ be pre-trade nominal holdings, $V_t^- = \sum_i h_{i,t}^-$, and $p_{i,t} = h_{i,t}^-/V_t^-$. A target w_t moves the following amount and incurs the following fee at rate c:

$$Q_t = \frac{V_t^-}{2}\|w_t - p_t\|_1, \qquad F_t = cQ_t, \qquad h_{i,t}^+ = (V_t^- - F_t)w_{i,t}. \tag{1}$$

The one-half factor counts the transferred money once, rather than charging separately for both sale and purchase. Turnover is measured against pre-fee wealth; the fee is deducted before returns, and the remaining wealth is allocated to the target. This states the accounting convention explicitly rather than assuming an implicit exact-dollar trade solver.

For nominal return $r_{i,t}$ and inflation π_t, define real return $\rho_{i,t}$, cumulative price index I_t, and year-end real wealth W_t by

$$\rho_{i,t} = \frac{1 + r_{i,t}}{1 + \pi_t} - 1, \qquad I_t = \prod_{k=1}^{t}(1 + \pi_k), \qquad W_t = \frac{\sum_i h_{i,t}^+(1 + r_{i,t})}{I_t}. \tag{2}$$

Pending human review means that existing assets remain invested and automatic trades stop. No unobserved human intervention, liquidation, or recovery is credited to the algorithm.

2.2 Growth, drawdown, and the cost of intervention

For a scenario with horizon H, real compound annual growth is

$$G = \left(\frac{W_H}{100}\right)^{1/H} - 1. \tag{3}$$

Maximum drawdown measures loss from a previous real-wealth peak. Let Z_j contain the initial wealth, each post-fee/pre-return wealth mark, and each year-end wealth mark. Then

$$D = \max_j \left(1 - \frac{Z_j}{\max_{k \leq j} Z_k}\right). \tag{4}$$

Including the post-fee marks ensures that trading losses are not hidden between annual observations. A larger D is worse. A stated customer tolerance L is evaluated through the event $D > L$; it is a preference threshold, not a guaranteed loss limit.

Across scenarios, we report median G, P10(G), median D, and P90(D). Annual volatility is the sample standard deviation of net real annual returns; its cross-scenario median summarizes typical variability. Terminal-loss probability measures $W_H < 100$. Drawdown CVaR at 90% averages the worst 10% of scenario drawdowns, with fractional boundary weights when needed; it is distinct from a 90th percentile.

The main trading frequency is the fraction of reviews in periods $2, \ldots, H$ that execute a trade. Initial allocation is excluded because it is compulsory; the alternative rate including it is reported when relevant. Mean real fees accumulate inflation-adjusted fees over the horizon, in units of initial wealth 100. Review incidence is the fraction of scenarios that ever request human review.

3 Strategic Portfolios and Customer Suitability

3.1 A convex allocation problem for each horizon

For N training paths and a fixed annually restored target w, define the largest cumulative log loss over any contiguous interval:

$$Y_s(w) = \max_{0 \le u \le v \le H} \left\{ - \sum_{t=u+1}^{v} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \right\}. \tag{5}$$

The empty interval contributes zero. Under this cost-free, fixed-target model, ordinary maximum drawdown satisfies $D_s = 1 - \exp(-Y_s)$. Tail expectations must still be calculated on the intended variable: transforming $\mathrm{ES}(Y)$ does not generally give $\mathrm{ES}(D)$.

Using the variational representation of expected shortfall [2],

$$\mathrm{ES}_\alpha(Y) = \min_z \left[z + \frac{1}{N(1-\alpha)} \sum_{s=1}^{N} (Y_s - z)_+ \right], \quad \alpha = 0.90, \tag{6}$$

we choose the strategic anchor w^* by solving

$$\min_w \quad - \frac{1}{NH} \sum_{s=1}^{N} \sum_{t=1}^{H} \log(1 + \rho_{s,t}^{\mathsf{T}} w) \tag{7}$$

$$\text{subject to} \quad \mathrm{ES}_{0.90}(Y(w)) \le q R_{\text{ref}}, \quad w \ge 0, \quad \mathbf{1}^{\mathsf{T}} w = 1. \tag{8}$$

Conservative additionally requires $w_{\text{bill}} \ge 0.40$. For each horizon, R_{ref} is the training log-drawdown expected shortfall of a reference allocation. The equity/bond/bill references are $(.20, .30, .50)$ for Conservative, $(.50, .35, .15)$ for Balanced, and $(.75, .20, .05)$ for Growth. Thus products differ through their risk budgets as well as the Conservative liquidity floor.

Negative log growth and the maximum of interval log losses are convex on the valid-return domain. Equation (8) is therefore a convex strategic subproblem. A Kelley cutting-plane linear-programming method uses supporting planes, a 5% simplex grid for initialization, and lower/upper objective bounds to assess convergence. The configured optimality-gap tolerance is 10^{-6}, risk tolerance is 10^{-8}, and iteration cap is 300. Infeasibility is recorded rather than repaired by silently relaxing the risk budget. These certificates concern the strategic subproblem; they do not prove global optimality of the later trading and escalation system.

3.2 Development selection before freezing

Nested validation uses five outer scenario folds and three inner folds, preserving complete paths. Outer training/validation sizes are 96/24; inner training/validation sizes are 64/32. We compare $q \in \{.75, .85, .95\}$ and execution curvature $\gamma \in \{4, 12\}$, using held-out development outcomes rather than training fit alone. This separation addresses selection bias from repeated model comparisons [3].

Inner feasibility checks cover strategic feasibility, pooled log-drawdown risk, growth relative to same-anchor annual rebalancing, and review frequency. The validation risk budget is fixed independently of the candidate q; the risk slack is 0.005. Permitted median/P10 annual-growth shortfalls are 0.001/0.002, and review incidence is capped at 10%. Feasible candidates maximize mean $\log(1 + G)$; a minimum-violation fallback is explicitly diagnostic. The selected full-development refit uses $q = 0.95$ and $\gamma = 4$ for all three products, then freezes the anchors and customer assignments before final evaluation.

3.3 Frozen allocations and an explicit refusal option

Table 1 shows that a product name does not imply one allocation for every customer. Shorter-horizon Balanced portfolios hold more bills, while the 30-year Growth solution reaches the boundary with zero bills. These are model outputs, not general investment prescriptions.

Table 1: Frozen strategic equity / bond / bill weights (%). Rounding can make totals differ slightly from 100.

H	Conservative	Balanced	Growth
5	24.48 / 4.46 / 71.05	54.26 / 0.00 / 45.74	75.21 / 0.00 / 24.79
6	24.22 / 1.59 / 74.20	53.25 / 4.46 / 42.29	74.61 / 0.00 / 25.39
15	22.67 / 7.23 / 70.10	48.65 / 21.78 / 29.58	71.38 / 23.59 / 5.03
25	25.98 / 2.84 / 71.18	49.78 / 18.43 / 31.79	71.24 / 24.35 / 4.41
30	26.48 / 2.85 / 70.67	50.81 / 14.44 / 34.75	70.91 / 29.09 / 0.00

For customer horizon H and tolerance L, we count out-of-fold development paths with $D > L$, using the complete execution policy. If k of $n = 120$ paths breach, the one-sided Clopper–Pearson upper bound is

$$U = \text{Beta}^{-1}(1 - \delta; k + 1, n - k), \qquad \delta = 0.05/440, \tag{9}$$

with $U = 1$ when $k = n$. The 440-comparison family is the prespecified service-screening budget. Automatic service requires $U \leq 0.20$ under both the base cost and doubled cost. Because fitted out-of-fold policies share training observations, this is a conservative screening diagnostic; it is not a strict coverage guarantee for the final refitted policy.

High-liquidity customers require a strategic bill weight of at least 40% in every relevant development fold. This condition is checked for any eligible product, rather than restricting high-liquidity customers to the product named Conservative. It is a target requirement, not a continuously maintained cash floor after market drift. Growth is available only for horizons of at least 15 years. We also check that P90 drawdown is ordered Conservative $\leq$ Balanced $\leq$ Growth at the relevant horizon and costs. Among eligible products, the highest out-of-fold median growth is selected; if none qualify, the customer goes to human review.

Table 2: Customer requirements and assignments frozen before final evaluation.

Customer	H	Tolerance (%)	Liquidity	Assignment
A	5	10.00	High	Human review
B	15	25.00	Medium	Conservative
C	30	40.00	Low	Conservative
D	25	15.00	Medium	Human review
E	6	35.00	High	Balanced

B and C qualify for Conservative: their base-cost breach counts are 3/120 and 0/120, with adjusted upper bounds 12.45% and 7.29%. C's Balanced alternative has an upper bound of 28.93%, exceeding the service ceiling despite a raw breach rate of 14.17%. E qualifies for Balanced at six years: its bound is 13.87% at both costs, and its minimum fold-level bill target is 40.03%. A and D fail even the Conservative risk screen, with base upper bounds of 39.70% and 65.17%. Human review is therefore a substantive outcome of the model.

4 Annual Execution and Human Escalation

4.1 Candidate trades and the cost gate

At each annual review, the advisor checks for escalation before considering a trade. Let p denote current portfolio weights and w^* the frozen strategic anchor for the investment horizon. The five candidates are

$$w_\lambda = p + \lambda(w^* - p), \qquad \lambda \in \{0, .25, .50, .75, 1\}. \tag{10}$$

Here, $\lambda = 0$ keeps holdings and $\lambda = 1$ restores the anchor. Training returns supply the mean μ and regularized covariance $\Sigma = 0.75S + 0.25\,\mathrm{diag}(S) + 10^{-10}I$. The tracking score and fractional trading cost are

$$A(w) = 1 + \mu^\mathsf{T} w^* - \frac{\gamma}{2}(w - w^*)^\mathsf{T}\Sigma(w - w^*), \qquad f_\lambda = \frac{c}{2}\|w_\lambda - p\|_1. \tag{11}$$

The return term is shared across candidates; the quadratic term penalizes departure from the anchor. If any score is nonpositive or nonfinite, the advisor requests human review. Otherwise, it selects j maximizing $(1-f_\lambda)A(w_\lambda)$ and accepts a trade only if

$$\frac{A(w_j) - A(p)}{A(w_j)} > f_j + 0.0001. \tag{12}$$

The extra 0.0001 is a one-basis-point safety margin. If the selected candidate fails this strict gate, holdings are kept; lower-ranked candidates are not tried. This score measures tracking benefit, not a guaranteed increase in future wealth.

4.2 Four observable actions

Before trading, the advisor compares observed real drawdown with the training-based threshold

$$\tau = \mathrm{clip}(1.5\ \mathrm{P90}(D_{\text{train,annual}}),\ 0.10,\ 0.75)\,, \tag{13}$$

where $D_{\text{train,annual}}$ comes from fee-paying annual rebalancing to the same anchor. For 30-year Balanced A, $\tau = 63.85\%$. Table 3 distinguishes the four actions. Escalation takes priority over candidate selection; while review is pending, the engine holds existing assets and suspends automatic trading.

Table 3: Annual action rules. Accepted trades are classified against the previously recorded execution target, not necessarily the strategic anchor.

Action	Decision rule
1. Keep holdings	The selected step is zero or fails Equation (12); no trade fee is charged.
2. Rebalance	The accepted trade restores the recorded execution target.
3. Change target	The accepted trade sets a different execution target; w^* remains fixed.
4. Request human review	Observed drawdown reaches τ, or a candidate score is invalid; existing assets remain invested.

4.3 Customer-facing explanation

Every accepted adjustment uses this 20-word explanation, within the required 50-word limit:

> "This adjustment moves your investments toward the strategic allocation. Its estimated risk-adjusted benefit exceeds trading costs. Investment losses remain possible."

5 Final Evaluation Results

5.1 Higher growth comes with greater drawdown

Table 4 compares policies on the same 60 final scenarios, charging 0.25% of money moved. Equal-weight annual rebalancing is the required baseline; same-anchor annual and hold controls isolate execution effects.

Table 4: Thirty-year final outcomes across 60 scenarios (%). G: net real CAGR; D: maximum real drawdown. Trade rate excludes initial allocation.

Policy	Med. G	P10 G	Med. D	P90 D	Trades
Equal-weight annual	3.38	1.50	22.30	39.53	100.00
Balanced A	4.15	1.64	30.58	48.02	35.34
Same-anchor annual	4.02	1.67	30.30	48.12	100.00
Same-anchor hold	4.27	1.42	36.05	59.15	0.00
Conservative A	2.76	1.35	19.19	31.91	28.45
Growth A	5.14	2.07	38.23	62.15	28.56

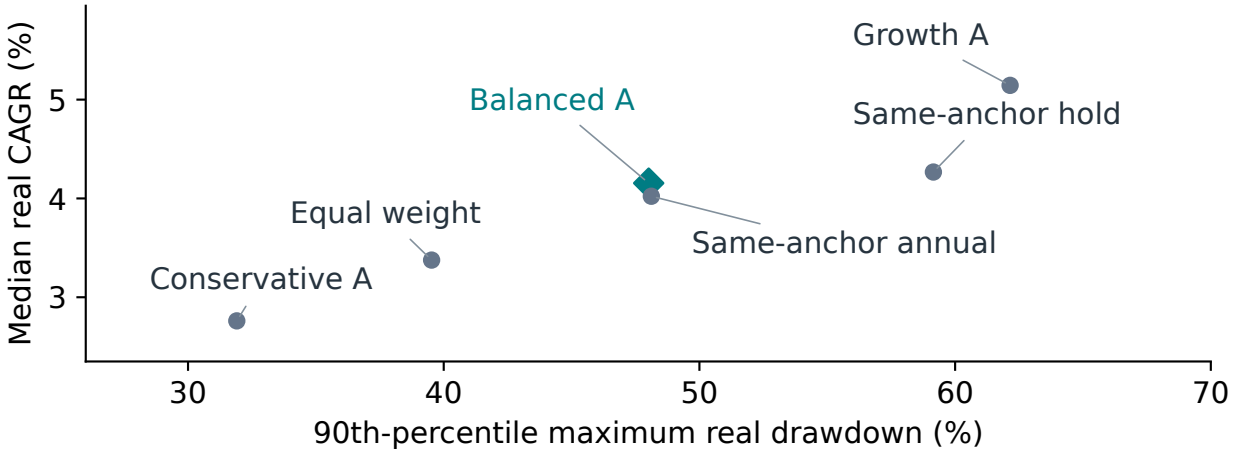


Figure 1: Thirty-year median growth versus P90 drawdown across 60 scenarios at the base fee. P90 is not a loss guarantee.

Balanced A gained 0.78 percentage points (pp) of median growth over equal weight, with 8.49 pp more P90 drawdown (Figure 1). Its median annual volatility was 10.76%, compared with 8.24% for equal weight.

Table 5: Paired bootstrap comparisons, Balanced A minus equal weight (pp); 4,000 complete-scenario resamples.

Metric	Difference	95% interval	Family-adjusted 95%
Median G	+0.78	[0.325, 1.184]	[0.098, 1.372]
P90 D	+8.49	[5.203, 13.365]	[3.399, 15.735]

Family adjustment covers 20 nonbaseline policies within each endpoint, horizon, and cost; it does not cover the whole adaptive research process.

Against same-anchor annual rebalancing, Balanced A gained 0.13 pp of median growth with nearly unchanged P90 drawdown. Mean real fees fell 26.83%, from 0.7973 to 0.5834. Its trade rate was 35.34% (37.50% including initial allocation). Relative to holding the anchor, P90 drawdown fell 11.13 pp and median growth fell 0.11 pp. These comparisons favour selective rebalancing when fees and tail losses both matter.

5.2 Customer outcomes at their actual horizons

Table 6 uses each customer's actual horizon and frozen assignment. A and D were referred for human review; dashes indicate no automated investment path. Breaches mean maximum drawdown exceeded the customer's stated tolerance.

Table 6: Final customer outcomes (growth and drawdown in %; 60 scenarios).

Customer	H	Med. G	P10 G	Med. D	P90 D	Breaches
A	5	–	–	–	–	–
B	15	2.94	0.48	12.57	25.38	7/60
C	30	2.76	1.35	19.19	31.91	3/60
D	25	–	–	–	–	–
E	6	4.11	-0.35	11.95	26.89	3/60

Table 7: Customer risk and trading. Terminal loss means final real wealth below 100; trade rates exclude initial allocation.

Customer	Median volatility (%)	Terminal losses	Trade rate (%)
B	6.26	3/60	22.86
C	6.75	3/60	28.45
E	10.08	9/60	30.67

For B, P90 drawdown exceeded the 25% tolerance, with breaches in 7 of 60 scenarios. Eligibility screens breach probability, rather than imposing a guaranteed loss limit. E's negative P10 growth and 9 terminal losses highlight the risk of its six-year horizon despite positive median growth. B, C, and E requested human review in 2, 3, and 0 scenarios, respectively. All assignments remained unchanged after final evaluation.

5.3 A severe path exposes the limits of escalation

In scenario 155, Balanced A peaked at 162.33 in year 6 and ended at 56.50 (Figure 2). Its 65.19% maximum drawdown and -1.88% real CAGR show the limits of annual escalation.

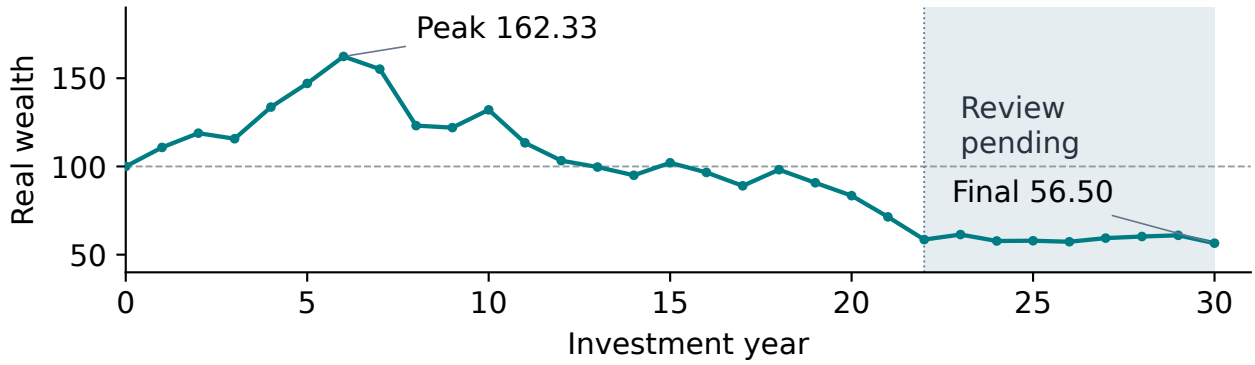


Figure 2: Saved real wealth for scenario 155, starting from 100. Lines join year-end observations; shading marks pending review, beginning before year 23.

Six post-initial trades preceded the review request before the year-23 return. Automatic trading then stopped, but the existing holdings remained exposed. Across all 60 Balanced paths, drawdown CVaR90 was 59.31%, versus P90 drawdown of 48.02%.

6 Sensitivity and Reliability

6.1 Transaction-cost stress with the policy frozen

The saved final cost sweep changes the fee rate while keeping the strategic model and customer assignments fixed. The cost gate still responds to the fee it faces. Table 8 and Figure 3 therefore describe a fixed decision rule under different execution costs, not a refitted strategy.

Table 8: Thirty-year Balanced A under alternative transaction costs. Growth and drawdown are in %; fees are real wealth units.

Cost (%)	Med. G	P10 G	P90 D	Mean fees	Trades (%)
0.000	4.08	1.68	47.96	0.0000	46.61
0.125	4.19	1.70	47.56	0.3199	39.60
0.250	4.15	1.64	48.02	0.5834	35.34
0.500	4.13	1.61	47.35	1.0668	28.68
1.000	4.13	1.61	47.72	1.8687	19.83

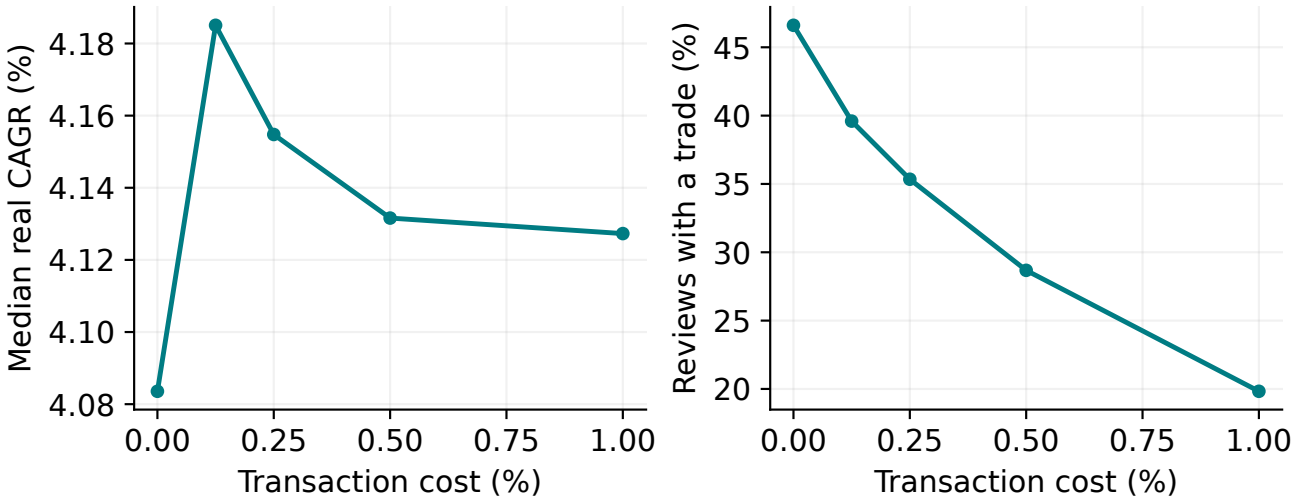


Figure 3: Higher fees suppress trading; growth need not fall monotonically because the actions also change.

Doubling the fee to 0.50% reduces the trading rate to 28.68%, while median real growth remains 4.13%. Mean real fees rise to 1.0668. Small nonmonotonic growth differences should not be interpreted as a benefit from charging fees: changed trades lead to changed exposures on the same finite set of scenarios.

6.2 What is verified, and what remains uncertain

The saved accounting audit passed all recorded checks, including 286,920 transition records, 780 scalar-reference cases, and all five customer assignments. Its maximum cross-policy return reconstruction discrepancy was 3.64×10^{-12}. The report-building audit checks frozen identity, source hashes, assignment consistency, and table extraction without reopening market data or reevaluating policies.

The strengths are consistent real-wealth accounting, scenario-level separation, and explainable suitability and trading rules. However, synthetic paths and only 60 final observations limit external and tail-risk conclusions. Annual observation misses within-year losses. Strategic risk excludes fees and differs from realized policy risk; bill weights can drift below targets. Human response, withdrawals, taxes, and additional frictions are unmodeled. The cost sweep tests fees, not these other assumptions. Improvements require a new development protocol and independent evaluation data.

7 Recommendation and Conclusions

We retain the frozen calibrated Plan A as the final submission strategy. Its three products use horizon-specific strategic allocations, and its execution layer trades only when a local benefit estimate clears costs and a safety margin. Customer screening is separate from return maximization: B and C receive Conservative, E receives Balanced, and A and D are referred to a human advisor.

The main empirical result is a trade-off. Balanced A improves 30-year median real growth over annual equal weight, but also produces greater tail drawdown. A fairer test of the execution layer compares the same anchor: selective trading cuts mean real fees by 26.83% relative to annual restoration, with similar tail drawdown in the final sample. Conversely, the severe path demonstrates that even an observed review trigger does not stop an ongoing loss.

SafeInvest should explain its targets, disclose losses in real purchasing-power terms, state that assets remain invested during review, and retain a refusal option. Neither the service ceiling nor the training risk constraint guarantees protection. Faster human response and stronger downside controls remain future research questions requiring independent evaluation.

Reproducibility and attribution

The implementation preserves the frozen model, development selection record, one-time final-evaluation lock, and saved comparator outputs. Tables and figures are generated directly from those records. Supplementary source code is a separate submission artifact; this report is self-contained.

The problem statement and supplied scenario data are credited to the 2026 MCM Senior Group organizers and their stated underlying JST source [1, 4]. We acknowledge the supplied materials' Creative Commons Attribution–NonCommercial–ShareAlike 4.0 license. Derived data displays in this report retain that attribution and are shared under the same terms. All returns in this report describe the supplied synthetic experiment, not forecasts of future asset performance.

References

[1] 2026 MCM Senior Group Organizers. *Problem A: SafeInvest: Building a Trustworthy Robo-Advisor*. Competition problem statement and supplied scenario data, 2026.

[2] R. T. Rockafellar and S. Uryasev. *Optimization of Conditional Value-at-Risk*. Author manuscript, 5 September 1999. https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf.

[3] G. C. Cawley and N. L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. https://www.jmlr.org/papers/v11/cawley10a.html.

[4] O. Jordà, M. Schularick, and A. M. Taylor. *JST Macrohistory Database*. Source series identified in the competition statement as Release 6. https://www.macrohistory.net/database/. Accessed 19 October 2026.

Use of AI Tools

OpenAI ChatGPT/Codex assisted with code development, analysis, drafting, and document checks. Numerical claims in this report are drawn from saved experiment outputs and frozen model settings; the report-building step did not refit a model or rerun the final evaluation. AI-generated explanations and prose require review by the named team members before submission. The authors remain responsible for the correctness, originality, attribution, and final submitted content.